

**ARSI UNIVERSITY
COLLEGE OF BUSINESS AND ECONOMICS
DEPARTMENT OF MANAGEMENT**

MODEL EXIT EXAM

Name: _____ **ID.**

Instruction: Choose the best answer and write the letter of your choice in the answer sheet (1 point each)

1. The measure of central tendency that is influenced most by extreme value is
 - A. Median
 - B. Mean**
 - C. Mode
 - D. The variance
2. The midpoint of the values after they have been ordered from smallest to the largest or from the largest to the smallest is called
 - A. Mean
 - B. Median**
 - C. Lower quartile
 - D. Upper quartile
3. A numerical value used as a summary measure for sample such as sample mean is known as
 - A. Population parameter
 - C. Sample statistic**
 - B. Sample parameter
 - D. Population mean
4. Statistical techniques that summarize and organize the data are classified as
 - A. Population statistics
 - C. Descriptive statistics**
 - B. Sample statistics
 - D. Inferential statistics
5. Which of the following sets of scores has the greatest variability?
 - A. 2, 5, 8, 11**
 - C. 20, 25, 26, 27
 - B. 13, 13, 13, 13
 - D. 42, 43, 44, 45
6. Which of the following statements is the most accurate description for the concept of Standard deviation?

- A. The total distance from the smallest score to the highest score.
 - B. The square root of the total distance from the smallest score to the highest score.
 - C. The squared average distance between all scores and the mean.**
 - D. The average distance between a score and the mean.
7. According to the empirical rule, the bell shaped distribution will have approximately 68% of the data within what number of standard deviation of the mean?
- A. One standard deviation**
 - B. Two standard deviation
 - C. Three standard deviation
 - D. Four standard deviation

The following data (**table 2.1**) shows the number of hours worked by 200 Arsi university management students

Class	Number of hours frequency
0 – 9	40
10 – 19	50
20 – 29	70
30 – 39	40

8. Refer **table 2.1** the class width for this distribution is
- A. 9
 - B. 10**
 - C. 11
 - D. Varies from class to class
9. Refer table 2.1 the number of students working 19 hours of less is
- A. 40
 - B. 50
 - C. 90**
 - D. 70
10. In normal distribution

A. Mean = Median = Mode

B. Mean > Median > Mode

C. Mean < Median < Mode

D. Mean $\neq$ Median $\neq$ Mode

Illustration: A manufacture of lawn and garden equipment makes two basic types of lawn mowers; a push type and a self-propelled model. The push type require 9 minutes to assemble and 2 minutes to package; the self-propelled mover requires 12 minute to assemble and 6 minutes to package. Each type has an engine. The company has 12hrs of assembly time available, 75 engines, and 5hrs of packing time profits are Birr 70 for the self-propelled model and br 45 for the push type mower per unit. Let X_1 and X_2 be push type mower and self-propelled mower respectively, (Answer Question 11 – 18 based on the above illustration)

11. What is the objective function?

A. $Z \min = 70X_1 + 45X_2$

C. $Z \max = 45X_1 + 70X_2$

B. $Z \max = 70X_1 + 45X_2$

D. $Z \min = 45X_1 + 70X_2$

12. How many operational constraints are there in the Model?

A. 3

B. 2

C. 4

D. 1

13. Which one is correct function/equation about assembly time constraint?

A. $2X_1 + 6X_2 \leq 5$

C. $9X_1 + 12X_2 \leq 12$

B. $9X_1 + 12X_2 \leq 720$

D. $0.15X_1 + 0.2X_2 \leq 12$

Calculate 2nd iteration of the above LPPM and answer questions 8 – 12 below.

14. What are the basic variables in the simplex table?

A. X_1, X_2 and S_3

C. S_1, X_1 and S_3

B. S_1, S_2 and S_3

D. S_1, X_2 and S_3

15. What is the entering and leaving variables in the simplex table respectively?

A. X_1 and S_3

X_1 and X_2

C. X_1 and S_1

D. X_2 and S_2

16. What is the value of pivot element and objective function in the simplex table respectively?

A. 5 and 3500

B. 1/3 and 50

C. 5 and 120

D. 6 and 0

17. What are the basic variables in the optimal solution?

A. S_1, X_2 and S_3

B. X_1, X_2 and S_3

C. X_1, S_2 and S_3

D. X_2, S_2 and S_3

E. None

18. What is the value of the objective function in the optimal solution?
- A. 42 B. 24 C. 4020 D. 9
19. Exceptions of minimization problems with the mixed maximization problem
- A. The coefficient of the artificial variable (M) is given a positive sign
B. The selection of the variable to enter the solution is based on the largest positive value in the C-Z row.
C. A solution is optimal when all the C - Z values are zero under the basic variables and negative numbers under non-basic variables.
D. The coefficient of the artificial variable (M) is given a negative sign
20. Analog models of operation research are:
- A. physical replications of real objects
B. Convey important physical characteristics
C. physical models that look like reality
D. physical models; but they substitute rather than replicate reality
21. Initial feasible solution of transportation is obtained when:
- A. The number of occupied cells equal one less than the sum of number of rows and number of columns in a transportation table
B. The number of occupied cells equal to the sum of number of rows and number of columns in a transportation table
C. The number of occupied cells is less than the sum of number of rows and number of columns in a transportation table
D. The number of occupied cells is more than the sum of number of rows and number of columns in a transportation table
22. One of the following is true about rules for tracing stepping – stone paths
- A. It is not permissible to skip over unoccupied cells to find an occupied cell from which the path can continue
B. A path moves starting and ending with the occupied cell that is being evaluated
C. Beginning with a + sign in the cell being evaluated.
D. We evaluate occupied cells for possibility of optimal transportation
23. One is correct about the MODI (Modified Distribution) method of evaluating a transportation solution for optimality
- A. Rows and columns index number determination is based on the unit costs of the unoccupied cells
B. When cell evaluations are positive or zero, an optimal solution has been found
C. The sum of the row index and the column index equals the cell's unit transportation cost of empty cell
D. The index numbers can be used to obtain the cell evaluations for occupied cells

24. In the case of the transportation problem, the existence of an alternative solution is signaled by:

- A. An empty cell's evaluation equal to negative
- B. An empty cell's evaluation equal to zero
- C. An empty cell's evaluation equal to positive
- D. An empty cell's evaluation equal to (number of columns) + (number of rows) – 1

Illustration: Suppose that a firm has three factories and four warehouses. The firm's production capacity at the factories and the demand for the four distribution centers (warehouses) located at various regions is given below.

Production Centers (Factories)	Production capacity (units)	Distribution or Destination centers	Demand forecast (units)
F ₁	1000	W ₁	1000
F ₂	2500	W ₂	500
F ₃	1500	W ₃	2000
		W ₄	1500

The cost of shipping each unit from the factories to ware houses through each route is given as follows.

Transport cost per unit

Origin	W ₁	W ₂	W ₃	W ₄
F ₁	Br 3	Br 2	Br 4	Br 6
F ₂	8	5	4	7
F ₃	10	9	11	12

Based on the above information answer question number 25 to 29

25. In an intuition approach of finding an initial feasible solution, which cell could be occupied first?

- A. F1 to W1 B. F3 to W1 C. F1 to W2 D. F3 to W3

26. In an intuition approach of finding an initial feasible solution, how much cost it incur a firm to transport from factory 2 to warehouse 3?

- A. Birr 3500 B. Birr 8000 C. Birr 5000 D. Birr 12000

27. How much is total cost of transportation in an intuition approach of finding an initial feasible solution?

- A. Birr 21,500 B. Birr 31,000 C. 5000 D. 13,500

28. How much is total cost of transportation in North West corner approach of finding an initial feasible solution?

- A. Birr 31,500 B. Birr 28, 500 C. Birr 5000 D. Birr 31,000

29. In North West corner approach of finding an initial feasible solution, how many units of product is to be transported from factory 3 to warehouse 4?

- A. 1000 units B. 2000 units C. 500 units **D. 1500 units**

30. One of the following is not true about assignment problem of management science

- A. **Each job can be assigned to two or more employee**
 B. Each employee must have one assigned job
 C. One of its objective is to minimize the total time or cost of doing the work
 D. The jobs be performed simultaneously

31. An optimal solution of assignment problem is possible if:

- A. **The number of lines needed to cover the zeros is equal to the number of rows**
 B. The number of occupied cells is less than the number of rows plus the number of columns minus one
 C. Cell evaluation results are all positive or zero
 D. The number of lines needed to cover the zeros is less than the number of rows

ABC manufacturing company must decide whether it should purchase component part from a supplier or manufacture the component parts at its plant. If demand is high, it would be to the company's advantage to manufacture the components. However, if demand is low, ABC's unit manufacturing cost would be high due to underutilization of equipment. The projected profit in thousands of birr for ABC's make or buy decision is shown below.

Decision alternatives	Demand		
	Low (S_1)	Medium (S_2)	High (S_3)
(d ₁) Manufacture the components	-20	40	100
(d ₂) Buy the components	10	45	70

The states of nature have the following probabilities. Probability of low demand = 0.35, probability of medium demand = 0.35 and probability of high demand = 0.3

Based on the above illustration, answer question number 32 - 38

32. What is the right decision you recommend ABC manufacturing company under the situation of Maximax?

- A. To buy the components
B. To Manufacture the components
 C. Neither to buy nor to manufacture the components
 D. Either to buy or to manufacture the components

33. The better row maximum value under the situation of Maxi-max is _____

- A. 10 **B. 100** C. 170 D. 70

34. The better row maximum value under the situation of Max-min is _____

- A. 20 B. 100 C. 70 **D. 10**

35. What is the right decision you recommend ABC manufacturing company under the situation of Mini-max?
- A. To buy the components
 - B. To Manufacture the components
 - C. Neither to buy nor to manufacture the components
 - D. **Either to buy or to manufacture the components**
36. The better average value under the situation of lap lace (insufficient reason criterion) is _
- A. 70 B. 40 **C. 41.67** D. 71.82
37. What is the expected monetary value of manufacturing and buying respectively under the situation of risk environment?
- A. **37 and 40.25** B. 40 and 41.67 C. 100 and 70 D. 30 and 30
38. What is the right decision you recommend ABC manufacturing company under the situation of lap lace and risk condition respectively?
- A. To buy and to manufacture C. To manufacture and to buy
 - B. **To buy and to buy** D. To manufacture and to manufacture
39. Which one of the following is not true about project network diagram?
- A. **Two activities can have the same head and the same tail**
 - B. Two activities can have the same head but different tails
 - C. Two activities can have the same tail but different heads.
 - D. Dummy activity doesn't have any time requirement
40. If the length of activities A-B-C-D-E is 25 weeks and the critical path time of the activities is 30 weeks, what is the slack time of these activities?
- A. 0 week B. 2 weeks **C. 5 weeks** D. -5 weeks
41. All of the following represent characteristics of management except:
- a. It is a social process
 - b. It is dynamic
 - c. It is concrete**
 - d. It is universal
42. Which of the following is true about levels of management?
- a. Middle managers are larger than top managers
 - b. Lower managers lead non manager supervisee
 - c. Middle managers are called product managers

- d. Top managers prepares intermediate range strategies and plans
43. Among Henry Fayol's 14 management principles, which one used the organization's manpower mode in one roadmap:
- a. Unity of command
 - b. Unity of direction
 - c. Scalar chain
 - d. Esprit de corps
44. Which of the following is different from the rest?
- a. Program
 - b. Procedure
 - c. Policies
 - d. rules
45. which of the following is false about decision making?
- a. Sometimes managers make poor decision due to lack of time
 - b. Decision making requires exhaustive identification of alternatives solutions
 - c. Developing alternatives comes before alternative evaluation
 - d. Problem diagnosis come before problem categorization
46. Which of the following is true about organizing?
- a. Formal organization formed in non-impersonal way
 - b. Informal organizations are visible on organizational structure
 - c. Division of labor refers breaking down a task into its most basic elements
 - d. Staff authority contains the authority to control specific activity in other units
47. Narrow span of management characterized by the following except:
- a. Flat organization structure
 - b. More communication between superiors and subordinates.
 - c. Centralized authority of managers
 - d. Managers are underutilized and have ample time
48. Reviewing applicants credential, giving exam and conducting an interview to get the right employee refers to:
- a. Recruitment process
 - b. Application blank
 - c. Selection process
 - d. Manpower planning
49. In formal organization managers gets authority because they mainly have _____ power:

- a. Expert power
 - b. Legitimate power
 - c. Coercive power
 - d. Reward power
50. Which of the following element of controlling not used to improve performance current activities or projects?
- a. Input assemble monitoring
 - b. Progress monitoring
 - c. Final evaluation
 - d. Midterm evaluation
51. Wants for specific products that are backed by ability and willingness to buy them is called:
- A. Exchange
 - B. Produce
 - C. Want
 - D. Demand
52. The kind of utility that exists when a product is readily accessible to potential customers is
- A. Information Utility
 - B. Form Utility
 - C. Time Utility
 - D. Place Utility
53. Which of the following statements is *not correct* about marketing philosophies?
- A. The major concern of production concept is to make products accessible at lower price.
 - B. Maintenance of long-term relationship with customers is the concern of sales concept.
 - C. Organizations which focus on product concept will make superior quality products.
 - D. The Selling Concept favors undertaking an aggressive selling and promotion effort
54. The systematic process of gathering, analyzing, supplying and applying information (both qualitative and quantitative) about the external market environment
- A. Market Research
 - B. Marketing Information system
 - C. Marketing Intelligence
 - D. Comparative Analysis

55. The type of pricing where an organization sets higher price first and then reduces when competition soars is
- A. Penetration pricing
 - B. **Skimming price**
 - C. Cost plus pricing
 - D. Markup price
56. One of the following **do not** characterize proprietor form of business. It is
- A. Owned by single individual
 - B. Suitable for establishing small and micro businesses
 - C. The form in which the owner enjoys all the benefits as well as bears all the risks.
 - D. **The form in which an entrepreneur can easily raise funds needed to establish and operate the business**
57. In _____ form of business, the liability is limited to the amount of shares owned by the owners.
- A. Proprietorship
 - B. Partnership
 - C. **Corporation**
 - D. Joint venture
58. One of the following cannot be considered as importance of MSEs in developing countries?
- A. Economical Use of Capital
 - B. **Less mobilization of Local Resources to protect environment**
 - C. Less Dependence on Foreign Capital
 - D. Shorter Gestation Period
59. One of the following **cannot** be the cause of business failure?
- A. Inadequate Financing
 - B. Neglect
 - C. **Good ethics**
 - D. Disaster
60. One of the following is **not under** the main supporting packages for MSEs development in Ethiopia.
- A. Awareness creation about the sector
 - B. Provision of legal services to form legal business enterprises
 - C. **Hiring the necessary professionals to the business.**
 - D. Providing Technical and business management training

61. Which one of the following strategy is different from the other?
- A. **Diversification**
 - B. Retrenchment
 - C. Divestiture
 - D. Liquidation
62. Forward integration is an effective strategy in all of the following conditions, except?
- A. If the organization competes in a growing industry
 - B. If Present distributors/retailers have high profit margins
 - C. **If present suppliers are expensive and unreliable.**
 - D. If going forward help to bring or sustain the company's competitive advantage.
63. Which one of the following strategy helps to bring economies of scale for an organization?
- A. Vertical Integration
 - B. **Horizontal integration**
 - C. Unrelated diversification
 - D. Defensive strategy.
64. Michael Porter's Five Generic Strategies, stated that firms' competitive advantage can be gained from_____?
- A. Two different bases, Internal and External factor Analysis
 - B. Two different bases, External factors (opportunity and threats)
 - C. **Three different bases, cost leadership, differentiation, and focus.**
 - D. Five different bases, porter's five force analysis.
65. A strategy implementation tactic, which enable the firm to get a position in the mind of the customer by entering in to the market prior to rival firms, is known as?
- A. Late mover Advantage
 - B. Outsourcing
 - C. Reshoring
 - D. **First mover advantage**
66. Which one of the following is **False** about SWOT- matrix?
- A. **It is a tool used in the input stage of strategy analysis framework.**
 - B. It helps to generate four types of strategies.
 - C. It does not show how to achieve a competitive advantage
 - D. It considers major or key internal and external factors only.

67. Which one of the following is **False** about Quantitative Strategic Planning Matrix (QSPM)?

- A. Use information only from the information stage of Analytical framework.
- B. Determine the relative attractiveness of feasible alternative actions (strategy).
- C. All internal and external factors from EFE-matrix and Internal factors from IFE-matrix will be considered.
- D. It needs the attractiveness score for every factors to be assigned.

68. Which one of the following statement is **true** about strategy implementation?

- A. It is more of a science than an art.
- B. It calls a responsibility shift from strategist to functional managers.
- C. The focus is on effectiveness.
- D. Requires good intuitive and analytical skills

69. Which one of the following statement is **False** about policy?

- A. Policies reduce the decision time.
- B. Policy is only important to guide decision making for recurring problems.
- C. Policies set boundaries on the kinds of administrative actions (strategies)
- D. It is a kind of single use plan.

70. Which one of the following is **not correct** about conflict in an organization?

- A. It is not always bad to have conflict in an organization.
- B. It needs to be managed well for successful implementation of a strategy.
- C. An absence of conflict assures successful implementation of a strategy.
- D. As long as there is change, conflict is inevitable every time.

71. Transformational Leadership include all of the following, **except**?

- A) Idealized influence
- B) Individualized consideration
- C) Contingent reward
- D) Inspirational motivation

72. A leadership theory that states different styles of leadership may be more appropriate for certain types of decision-making is known as; _____

- A) Great man Theory
- B) Contingency Theory
- C) Trait Theory
- D) Behavioral Theory

73. Which of the following is/are **not true** about comparison of leadership and management?

- A) Management produces order and consistency; whereas leadership produces change and movement
- B) Management provides structure; whereas leadership communicates goals
- C) Management building team; whereas leadership provides rules and procedures
- D) A and C

74. Which of these best describes **restraining forces** to change?

- A) Negotiation
- B) Resistance to change
- C) Refreezing
- D) Unfreezing

75. The conflict between Management and Accounting department on resource usage is classified under which type of conflict?

- A) Intergroup conflict
- B) Intragroup conflict
- C) Interpersonal conflict
- D) Intra organizational conflict

76. All conflict should be approached with the ultimate goal of _____.

- A) Eliminating conflict from the workplace completely
- B) Reducing interpersonal conflict but encouraging task conflict
- C) Only having conflict that ends well
- D) Creating functional conflict while minimizing dysfunctional conflict

77. Which of the following mismatch?

- A) Autocratic style of leadership-----I approach
- B) Democratic style of leadership---We approach
- C) Transformational style of leadership---Cost-benefit exchange
- D) Laissez-Faire Leadership Style----Hands-off

78. A conflict management strategy in which two or more parties exchange goods or services and attempt to agree on the exchange rate of for them is known as_____.

- A) Negotiation
 - B) **Compromise**
 - C) Confrontation
 - D) Avoidance
79. Change can be originated from within an organization. Which of the following refers to this statement?
- A) **New corporate** leadership
 - B) Political conditions of a country
 - C) Global technological advancement
 - D) New government policy
80. A leadership model that involves leaders clarifying goals and objectives, communicating to organize tasks and activities with the cooperation of their employees to ensure that wider organizational goals are met is known as _____.
- A) Transformational leadership
 - B) **Transactional leadership**
 - C) Charismatic Leadership
 - D) Trait
81. What is the main interest of long term creditors on the financial ratio analysis of a company?
- A. The level of risk and return
 - B. Liquidity and profitability
 - C. Monitoring the performance of the company
 - D. **To evaluate the solvency position of the firm**
82. Suppose that you want to compare the financial performance of your business with other similar firm which is the most competitor in the industry, which method would you prefer as a standard of comparison (bench mark)?
- A. Trend analysis
 - B. Performa analysis
 - C. Time series analysis
 - D. **Cross sectional analysis**
83. What do we mean when we say the Standard deviation of an investment is 15%?

- A. An investor has 15% chance of realizing its expected return.
 - B. An investor has 15% chance of losing its expected return.
 - C. The possible outcome of the investment will disperse around the expected out come by 15%
 - D. The actual return of the investment is lower than the expected return by 15%
84. What is the main interest of long term creditors on the financial ratio analysis of a company?
- E. The level of risk and return
 - F. Liquidity and profitability
 - G. Monitoring the performance of the company
 - H. To evaluate the solvency position of the firm
85. One of the following is not function of financial market
- A. Borrowing and Lending
 - B. Risk Sharing
 - C. Information Aggregation and Coordination
 - D. Asset transformation
86. Secondary market is
- A. Market for short term instrument
 - B. Always an OTC market
 - C. Market in which investors resold their security
 - D. Market in which IPO is traded
87. Which one of the following is incorrect about the concept of cost of capital?
- A. If the rate of return of an investment is more than its cost of capital, the investment will have a positive NPV.
 - B. If the investment has a positive NPV, its its cost of capital will be more than the internal rate of return.
 - C. The cost of capital depends principally on the use of the funds, not the source.
 - D. Implementation of capital project with a rate of return more than cost of capital will maximize the wealth of the firm
88. The phase of a project study that embraces the period from the decision to invest to the start of commercial production is called

- A. Feasibility study
- B. Financial analysis
- C. Implementation
- D. Economic analysis

89. If the acceptance of one does not eliminate the other (s) from consideration, the project is/are called?

- A. Mutually exclusive
- B. Contingent investment
- C. Independent project
- D. Capital rationing

90. The following are components of the initial cost of a project except

- A. Building and civil works
- B. Cost of production
- C. Expenditures for preparatory studies
- D. Preliminary Expenditures

91. Which of the following is not a task in project identification and formulation?

- A. Idea generation
- B. Project implementation
- C. Preliminary screening
- D. Project evaluation

92. Which of the following is **correct**?

- A. A program is part of a project
- B. Unlike projects, a program is necessarily time bounded
- C. Unlike a program, projects have specific beneficiaries
- D. A project and a program are not related at all

93. Which of the following are factors to be considered in analyzing reasonableness of cost of production, except?

- A. Cost of material inputs
- B. Labor costs
- C. Factory overheads
- D. Sunk costs

94. The following aspects of a project study fall under technical analysis **except**

A. Raw material and supply study

B. Cost estimates

C. Choice of Technology

D. Capacity utilization

95. Technical analysis is not mainly concerned with

A. Ensuring sustainable supply for raw – materials

B. Determining optimal plant capacity

C. Determining purchasing power of consumers

D. Selecting suitable plant location